

TO WHAT EXTENT DOES MARKET CATEGORIZATION AND VOLATILITY INFLUENCE THE DEGREE OF PRICE INEFFICIENCY IN PREDICTION MARKETS?

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1. INITIAL PROBLEM DESCRIPTION

Prediction markets are widely regarded as efficient mechanisms for aggregating dispersed information and producing probabilistic forecasts of real-world events. In this context, market efficiency refers to the degree to which prices reflect all available information; in a perfectly efficient system, the cost of "Yes" and "No" contracts should sum to the price of the payout, or exactly \$1 [6]. However, behavioral factors such as herd mentality and friction in the release of information can skew these probabilities. While prior research has shown that prediction markets are generally well-calibrated and efficient [10, 3], less attention has been given to the mechanisms and conditions that drive inefficiencies.

The project initially focused on arbitrage, the most direct indicator of market inefficiency. Arbitrage is defined as a risk-free profit opportunity resulting from price discrepancies, making it a natural indicator of market inefficiency [1]. Specifically, if the combined cost of a "yes" and "no" contract is less than \$1, a trader can make a profit regardless of the outcome. In theory, these opportunities should be rare and short-lived, as rational traders who spot them will quickly exploit them and eliminate them. However, in practice, arbitrage may arise due to delays in information release, limited market liquidity, or structural constraints.

In the final iteration of my study, I wanted to identify when arbitrage opportunities occur to determine whether certain kinds of markets (like sports, politics, etc.) are more prone to such inefficiencies. Using data from the Polymarket API, which serves as a decentralized prediction market platform, this project aimed to analyze whether market category affects the likelihood of arbitrage. My final research question was "To what extent does market categorization and volatility influence the degree of price inefficiency in prediction markets?"

2. INITIAL PROOF OF CONCEPT (INCLUDING FALL TERM)

The fall term mostly consisted of collecting available data in the API and fine-tuning my question into something that could be answered with the data available. I developed a WebSocket connection to the Polymarket API, which could be used to gather real-time data. Over several days of continuous data collection, a large data set was collected, enabling preliminary analysis of the market dynamics. 1,159,154 total rows of data were collected, showing updates of 294 different markets. Each row contained bid and ask prices, order book updates, trade activity, timestamps, and other features. Developing the WebSocket and pulling data was the most time-consuming aspect of this project, as Polymarket frequently changed its policies regarding pulling data and trading fees [8]. Both the amount of data in terms of markets and the information on each bid were limited.

By December, I had collected the data I was planning on using for my analysis. The next steps, which continued into the Spring Semester, included data processing and cleaning to ensure that the data was complete and remained intact. To further expand on what I mean

by "intact", this included ensuring the probabilities were reasonable and that the data I was collecting was real and didn't include any unreasonable values.

I also did some brief analysis of the data and noticed that differences between prices were very small in magnitude and short-lived. It would be very difficult to actually make any profit off of these arbitrage opportunities with the limited liquidity and time given to act. Despite the limited presence of arbitrage, the initial proof of concept in the fall gave me valuable insight into how prediction markets are structured and how I wanted to structure my project. While I may not be able to focus on the short-lived and few instances of arbitrage, the data could still be used to identify market inefficiencies. The data also demonstrated that while markets are generally efficient, there exists variability in price behavior across different categories of markets.

This phase demonstrated the challenges associated with working with high-frequency financial data, as the data contained many missing values, was very noisy, and was nearly impossible to pull. Polymarket did not want to allow easy access to data on the API, making the fall semester a very rigorous process for collecting data and cleaning it. However, I was able to get a clear idea of how prediction markets function and fine-tune my research question into something that could be answered.

3. PROBLEM REVISIONS

As this project progressed and I was made aware of what prediction market data actually yielded, I was faced with many problem revisions based on what would be feasible to answer and what would yield interesting results.

Initially, as a full group, we wanted to look at whether prediction markets were actually accurate, and whether it would be possible to detect when insider trading occurred at a specific instant. These questions failed to be the center of my project for two reasons. First, prominent literature like the work of Surowiecki [9] and Wolfers and Zitzewitz [10] propose that these markets are efficient aggregators of information, but verifying "accuracy" numerically can be hindered by behavioral noise. The second is that any questions regarding insider trading would be impossible to answer. Market participants rarely disclose involvement in illicit activities, and there are implications to incorrectly accusing people of insider trading. Because my groupmates were set on exploring these aspects more in depth, I decided to introduce a problem that I would have an easier time answering.

I then adjusted the scope of my project to look at instances of arbitrage, as I knew this would be a good way to actually analyze market inefficiency (arbitrage only occurs when markets are inefficient). However, I faced issues with exclusively looking at instances of arbitrage, as very few instances occurred within one platform (Polymarket), and the data was highly imbalanced when simply assigning some rows to 0 for "not arbitrage" and 1 for "arbitrage". To address this limitation, I focused more on defining market inefficiency as a continuing variable rather than a binary one. I decided to introduce a continuous inefficiency metric defined as the deviation from pricing parity. Specifically, inefficiency was measured as the difference between one and the sum of the "yes" and "no" contract prices.

This shift allowed my project to have a more flexible and informative framework. Rather than simply asking whether arbitrage occurs, the revised framework looked at how close markets come to arbitrage opportunities and how this varies across time and market conditions. This reframing also allowed the use of regression-based machine learning models and allowed for a deeper look into which variables typically drive market inefficiency. I was

also able to emphasize market microstructure and focus on how features such as volatility, liquidity, and trading behavior influence price formation. By moving away from a binary classification problem and toward a continuous prediction task, the project became better aligned with both the available data and the underlying economic questions.

4. EXPERIMENTAL DESIGN

4.1. Data Collection and Processing. My data was collected from the Polymarket API using a continuous live WebSocket to pull data on current markets. As a result, I collected 1,159,154 total rows of data of 294 different markets. This included an average of 4,000 rows of bid, ask, and spread prices per market. While I was pulling the data, I actively sorted each market into a market category between sports, politics, geopolitics, crypto, and pop culture. This included using a word match to a dictionary I created myself based on what keywords were relevant in the current markets. I used one-hot encoding to ensure that the categories were compatible with learning algorithms that require numerical input and to provide a more detailed representation of each category. Because I was limited to markets that were currently open, the amount of data collected in each category varied. For sports, there were 703,003 rows, for geopolitics, there were 1,444 rows, for politics, there were 373,248 rows, for pop culture, there were 63,680 rows, and for crypto, there were 11,509 rows. Because there was so little market data for geopolitics, I decided to combine these rows with regular politics. For reference, the column information contained in each row is stated below:

```
Metadata:
category_final, end_date, event_type, market_id, no_token_id, question, slug,
start_date, timestamp_utc, yes_token_id

Features:
buy_both_cost, distance_from_parity, imbalance_no, imbalance_yes,
last_trade_price, last_trade_side_token, liquidity_num, mid_sum, no_ask,
no_ask_size, no_bid, no_bid_size, no_mid, sell_both_revenue, spread_no,
spread_total, spread_yes, volume_24hr, yes_ask, yes_ask_size, yes_bid,
yes_bid_size, yes_mid

Targets:
edge, sell_edge

Labels:
buy_both_arb, near_arb, sell_both_arb
```

FIGURE 1. Features Extracted from PolyMarket API Live WebSocket

The dataframe was then organized by market ID and time, allowing for ordered trades in each market to be analyzed. I imputed the NA values that remained in the independent feature columns by forward filling the yes bid and replacing the NA last trade prices with the mid price. I chose these methods to preserve the integrity of the limit order book without introducing forward-looking bias or artificial volatility. Forward-filling the "yes" bid assumes that the last known offer remains valid until a new one is published, which reflects market microstructure where an order stays in the book until it is filled or canceled. Replacing missing last trade prices with the mid-price provides a neutral estimate of the fair market value at that instant. Because mid-price is derived from the existing spread rather than future data points, imputing this value avoids creating price jumps or artificial trends that mean-imputation may introduce.

To allow for a more in-depth analysis of each market, I computed more columns that could be used as features in the model training. The first was volatility per market, which is a rolling standard deviation of the last ten returns within each market. A higher value would signify a market is more chaotic as of late, whereas a low value would signify a market

that is stable with little movement. I used the same idea to calculate spread volatility and observe the spread over the last 10 observations. I also calculated the spread change from the previous observation, and the mid return, which represented a percentage change in mid price from the previous observation. Finally, I calculated the target variable for my model, which was an inefficiency score that I defined as $1 - [\text{yes ask price} + \text{no ask price}]$. The higher this value is, the more inefficient and closer to arbitrage the market is. This allowed me to develop a model that would output a score, rather than a binary outcome.

4.2. Feature Engineering and Model Approach. As established in recent financial literature [4, 2], ensemble methods like random forests and gradient boosting are particularly effective at capturing non-linear risk-return trade-offs present in high-frequency data. I attempted to train a logistic regression model, and received very poor prediction results even with feature engineering. This could be because a logistic regression model attempts to draw a flat decision boundary, and the data has very intricate and complex feature interactions with volatility spikes. The data is large and messy, and random forest regressors are more robust to noise.

For the first step of testing, I observed the collinearity of the variables to see which I could exclude in the training of the model. For reference, I looked for a Variance Inflation Factor(VIF) rating below three[5], as values above this threshold indicate significant multicollinearity that can destabilize coefficient estimates. I also emitted columns in the data that would make the results of the model arbitrary (like yes ask price and no ask price, which were used to create the target), and I was left with seven feature columns: spread volatility, rolling volatility, category: pop culture, category: sports, category: politics, mid return, and spread change (category: crypto was left out, as this can be assumed when all other categories are 0).

50% of the rows were assigned to the training set (652,101 rows), 25% were assigned to the validation set (326,050) and the remaining 25% were assigned to the testing set. Because the data is disproportionate in the number of rows assigned per category, I used stratified sampling to ensure that each set maintained identical category proportions. While a single stratified split was used in the final evaluation due to the computational intensity of the high-frequency dataset, I addressed the issue of resampling by performing 5-fold cross-validation during the hyperparameter tuning phase.

To optimize the random tree regressor model, I used a grid search with tuned hyperparameters to fit the model to the training data and optimize it on the validation data. While random tree regressors are not usually prone to overfitting, I examined 7 different hyperparameters, so testing was limited to a few hours. I set bootstrap equal to true in every search, as models always performed better with this hyperparameter. The following grid was used for grid search:

```
param_grid = {
    'n_estimators': [200, 400],
    'max_depth': [5, 10, None],
    'min_samples_split': [2, 5, 10, 20],
    'min_samples_leaf': [1, 2, 5],
    'max_features': ['sqrt', 0.5, 1.0],
    'bootstrap': [True],
    'max_samples': [0.5, 0.8]
}
```

FIGURE 2. Parameter Grid Using 5-fold Cross Validation

Hyperparameters for the final model were selected using 5-fold cross-validation over the parameter grid, with RMSE as the primary selection metric. As I ran this code locally, the process of using grid search with my selected values took about 5 hours when using the entire data set. To limit down which variables I wanted to use in grid search, I started with a much more exhaustive list that I tested on a much smaller set of data. In total, I examined 432 possible models on the full set of data.

5. EXPERIMENTAL RESULTS

5.1. Analyzing the Data. Before fitting any predictive models, I examined the structure of the overall data along with the data at the market category level. I analyzed four key market features, including total spread, log liquidity, log volume, and distance from parity. I graphed the results on the plots in Figure 3.

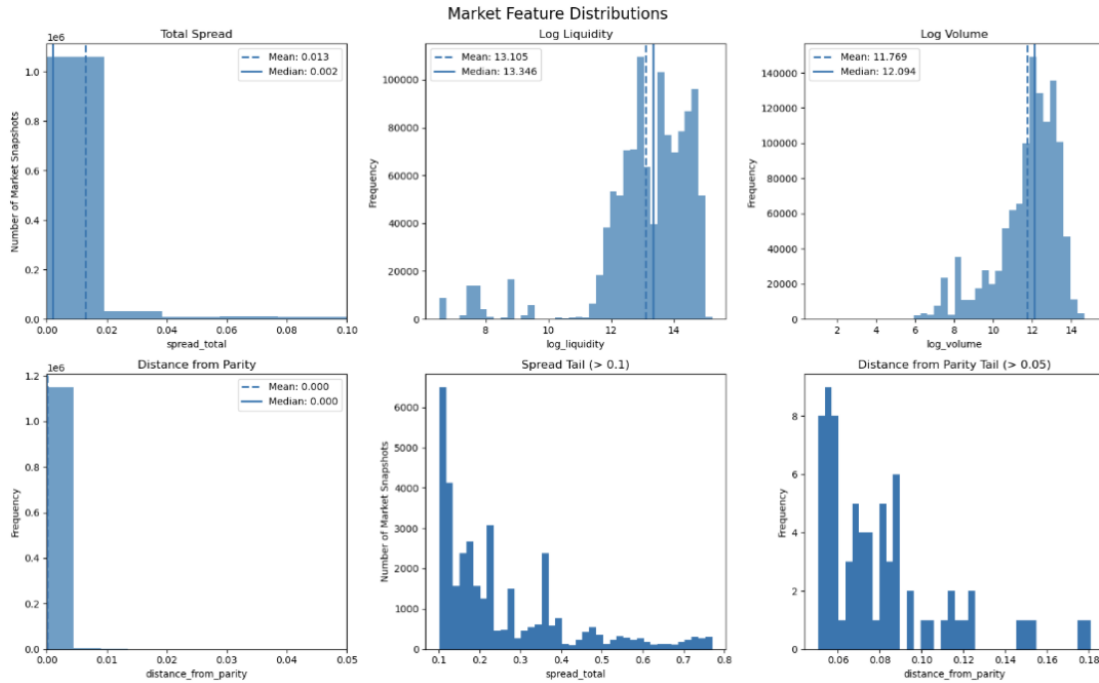


FIGURE 3. Spread, Log Liquidity, Log Volume, and Distance From Parity Distributions

The distribution of total spread across all markets is highly concentrated at 0, with a small right tail extending to about 0.7. Because efficient markets are high in volume and represent most of the data, this table could be misleading and minimize the amount of inefficiencies there are in the spread. After applying a logarithmic transformation to the liquidity, the liquidity seems to have a slight bell-curve shape with a longer left tail. Most markets are concentrated in a relatively narrow range from 12 to 14, corresponding to approximately \$160,000 to \$1.2 million in 24-hour trading volume. The similarity between the mean and median demonstrates a fairly symmetric distribution, indicating that extreme liquidity values are sparse. The log-transformed volume has a similar shape, with a slight bell-curve and long left tail. Most values are concentrated between 10 and 14, indicating that most of the markets in the data experience moderate to high levels of trading, with few markets exhibiting extremely high or extremely low volume. The distribution for distance from parity is almost entirely concentrated at 0, with very few deviations. Polymarket seems

to be generally well-calibrated, as the strict arbitrage values where prices deviate from parity is extremely rare.

I then examined the difference in spreads across the four different market categories in the plots below:

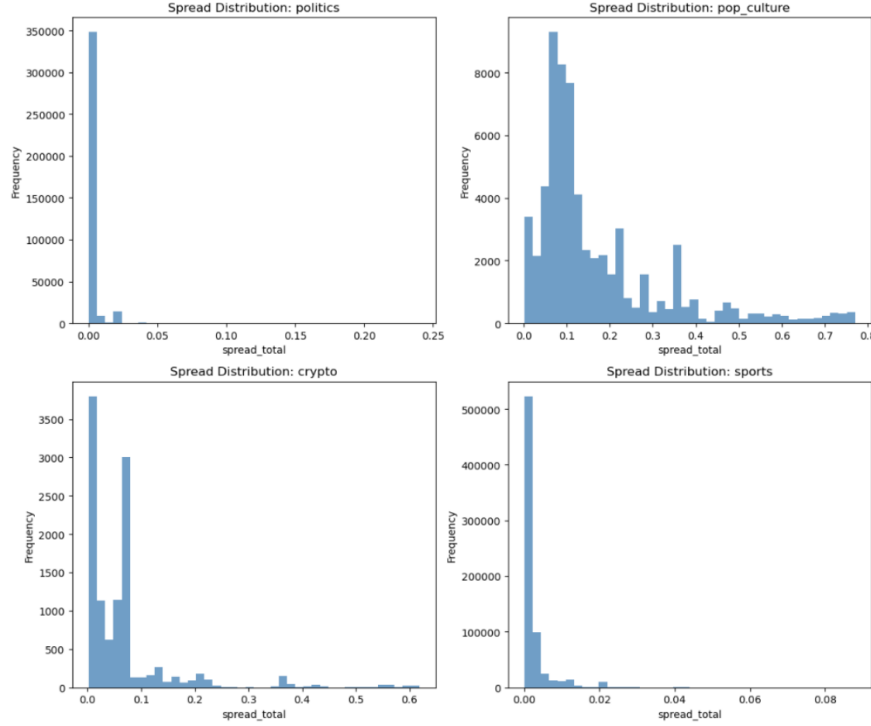


FIGURE 4. Spread Distribution Across Political, Pop Culture, Crypto, and Sports Markets

The spread distribution across political markets is highly concentrated around 0, with very few observations exhibiting slightly larger spreads. This indicates that these markets are the most efficient, with tight bid-ask spreads and minimal pricing discrepancies. Sports markets exhibit very similar behavior to political markets, where most spreads are concentrated around 0. I suspect political and sports markets are the most efficient due to public information access and high active trading in voluminous markets. The spread distribution for crypto is much more dispersed, with most values concentrated between 0 and 0.1 and a noticeable right tail. The presence of larger spreads demonstrates that crypto markets have more periods of reduced efficiency, likely due to more periods of volatility and price changes. Pop culture markets exhibit the widest spread distribution, with a substantial right tail and spread values up to 0.8. These markets are the most inefficient, most likely due to lower liquidity and slower-moving information. These values for spread seemed very extreme, so I more closely examined the data integrity of crypto and pop culture markets and found that the data was correct; these are just very inefficient markets. These patterns align with previous findings on anomalies in probability estimates for event forecasting, where certain market conditions can lead to persistent mispricing [7].

5.2. Performance of Model. The optimal random forest regressor model had parameters set to $n_estimators = 400$, $max_depth = 5$, $min_samples_split = 20$, $min_samples_leaf = 5$, $max_features = 1.0$, $bootstrap = \text{true}$, and $max_samples = 0.5$.

Below is the performance of the model on the training, validation, and test sets:

Dataset	Std Dev	RMSE	R^2	Normalized RMSE
Train	0.02931	0.01530	0.7266	0.2615
Validation	0.01059	0.00765	0.4757	0.7241
Test	0.03141	0.01952	0.6122	0.6227

TABLE 1. Model Performance Across Training, Validation, and Test Datasets

I chose to use Normalized RMSE as a performance metric, as the standard deviations and scale of each dataset were very different. The RMSE values are small relative to the standard deviation of the target value, indicating that the model captures a meaningful portion of the variation in market inefficiency. Possibly due to the temporal effects or differences in market conditions across the three sets, the model performs better on the test set than on the validation set. The test set may have a similar distribution to the train set, as this model is optimized to the validation set but performs best on the train and test sets. Across all sets, the r^2 value indicates that the model explains a moderate portion of the variation in efficiency, with the best performance on the test set. Prediction market data is very noisy, so even an r^2 value of about 0.5 shows a decent amount of predictive power from the model [4].

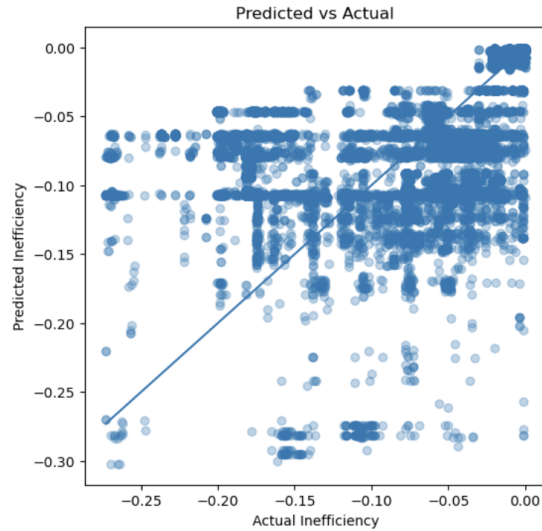


FIGURE 5. Predicted vs. Actual Inefficiency on the Test Set

The predicted vs. actual plot shows that the model captures general trends in inefficiency as shown by the upward trend of the data along the line. One downfall of the model is that it seems to underestimate extreme inefficiency values. From an actual inefficiency of -0.25 to -0.20, most of the predicted inefficiency values are above the line. The model seems to be better at predicting typical market conditions rather than rare, high inefficiency events. This could be due to the data containing a large amount of normal market conditions to a smaller amount of rare and extreme market conditions

	feature	importance
5	category_final_pop_culture	0.412528
1	spread_volatility	0.243715
2	volatility_10	0.148804
0	spread_change	0.079918
3	mid_return	0.048076
4	category_final_sports	0.040227
6	category_final_politics	0.026732

FIGURE 6. Feature Importance in Optimized Model

The most important feature in the model is the binary indicator of whether or not a market is categorized as pop culture at over 40% of the total feature importance. This model was trained to interpret pop culture markets as the least efficient markets, as this is the primary indicator of inefficiency. This result is consistent with the earlier spread analysis of the data, which showed that pop culture markets demonstrated the widest spread distributions. The model, therefore, relies heavily on categorical differences to predict inefficiency. The volatility-based features make up about 38% of the feature importance, showing that short-term fluctuations in market behavior are a strong predictor for inefficiency, with more volatile markets prone to larger price discrepancies.

6. CRITICAL ANALYSIS OF RESULTS. WHAT DO YOUR FINDINGS MEAN? HOW DO THEY INFORM FUTURE ACTION AND/OR RESEARCH?

The findings of this study indicate that inefficiency is not random, but instead systematically related to observable characteristics within the data. As inefficiency can be modeled somewhat accurately, the data contained in the API is not complete noise to the model and can be used to create an inefficiency score. Market category emerged as a dominant factor, showing how structural differences play a part in shaping price dynamics. This result is consistent with the pre-model spread analysis of the data, which showed that spreads within different categories varied significantly. While it is not a very interesting result, this data also showed that markets with higher liquidity and active participation tend to be more efficient, while markets with lower liquidity and low liquidity markets exhibit much greater variability.

The importance of volatility-related features suggests that short-term fluctuations in trading behavior contribute to temporary inefficiencies. These fluctuations can be caused by new information, a change in trader sentiment, or noise-driven trading activity.

Despite these insights, the model's performance in predicting extreme inefficiencies was the most limited. This was most likely due to the rarity of these events, as the model tends to play it safe and assume that a market is more efficient than it is. This is a huge limit to the model, as only inefficient markets can be taken advantage of, and only predicting typical market data does not allow it to be useful in those situations. The results of this study support the view that prediction markets are generally efficient but subject to inefficiencies driven by structural and behavioral factors of a market.

With access to more prediction market data, this project could inform traders on which markets are preferable to trade in. Generally, it is much lower risk to trade in markets that are efficient and less susceptible to price jumps. Because Polymarket allows everyone access to their WebSocket, there can be further developed models that could actually inform a trader on whether the market they want to trade in is "safe". My project was important

in determining that these inefficiencies are not random, but instead vary in frequency and amount depending on market type and other calculated volatility features.

7. DEAD ENDS AND/OR FAILED APPROACHES

One of the primary challenges encountered in this project was my focus on arbitrage detection. I made the decision to shape my whole project around this, which made it difficult to pivot when I was able to see that my data would not allow for a feasible arbitrage detection model. As most data science projects go, the direction of this project was consistently changing, and it was hard to detach myself from the problem when it would not work out. I was definitely stuck at this point, and it took a closer look at some other predefined models that already exist with prediction markets to decide that I wanted to use a regression model rather than a classification model. After my models performed very poorly on classification models, they finally produced some meaningful results when I implemented this change.

The largest issue I had with the final version of my project was that the markets that were pulled were very disproportionate. There were significantly fewer markets available to analyze at the time in crypto and pop culture. Because I used a WebSocket to examine live markets, I was only offered bid-ask data that was current. This is the only way to pull specific market data from the PolyMarket API, as any information on past markets that is available does not include bid-ask information. I tried to fix this issue by limiting the amount of sports and politics market data that was being pulled, but this did not account for the fact that there were so few pop culture and crypto markets available.

Another unsuccessful approach involved incorporating cross-platform data to identify arbitrage opportunities between different prediction markets. Differences in data formats and the lack of a reliable method for matching equivalent markets across platforms made this approach infeasible. Going forward, if I were able to resolve this issue, I would have many more recorded instances of arbitrage and higher rates of inefficiencies as these tend to occur more often across platforms.

8. LESSONS LEARNED

This project provided several important lessons about data science research, working with real-world financial data, and working in a group setting.

- (1) Data quality and pre-processing are critical, as the raw data initially contained noise, missing values, and inconsistencies that significantly impact results. I had to go through a few iterations of pre-processing, as it seemed that the first few times, I had missed inconsistencies or values that tampered with my results.
- (2) The importance of refining my research question became evident. Initial questions that focused on accuracy and insider trading were infeasible to answer. After I had shifted my plans to look at market inefficiency and microstructure, I had to change my ideas for the project yet again after being presented with the available data.
- (3) This project demonstrated that simple features like market category and volatility can provide significant explanatory power. Utilizing and understanding the underlying system and its structure was more valuable to the model than trying more complex models.
- (4) Working in a group is often very difficult. I think the three of us had different expectations for what we wanted this project to be, and often disagreed on the

direction we wanted to take the project. Maybe stating these ideas more clearly in the beginning of this capstone could have led to this project ending much differently.

9. OPEN PROBLEMS FOR FUTURE WORK

Future Research could extend this work in several directions. One promising area, as I mentioned above, is the integration of cross-platform data, which could increase the accuracy of the model and improve model robustness. With this data, more meaningful conclusions could be made about the behavior of prediction markets in general rather than just Polymarket.

Furthermore, while 5-fold cross-validation provided internal robustness for this project, future work should employ repeated random resampling of the training and testing sets. I was limited in this project by local computation constraints, but repeated random resampling would provide a more precise confidence interval for the model’s performance metrics.

Another direction is to make the market categories more granular, revealing more detailed patterns in inefficiency. This would include the purchasing of more data or further developing the WebSocket to allow for more data pulling. As the categories were very skewed, having more markets available and more specific categories would allow definitive observations to be made of each market.

It would also be very useful to use market inefficiencies as a gauge for prediction market users, allowing for them to make more informed bets. I don’t see any prediction markets actually implementing this, as the main goal of these markets is to take users’ money rather than focus on their well-being. However, it would be useful to state to prediction market users which markets are more likely to lose their money due to their microstructure, and not just the probabilities that are presented to them.

I believe this project only touched the surface of prediction market microstructure, but utilizing patterns in these markets can allow users to have more transparency in what they are betting on and the dangers of the market.

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